🚀 ResilioChain: Autonomous Supply Chain Exception & Triage

Solution Architecture & Technical Design Document

1. Executive Summary

In modern global logistics, exception management is the critical bottleneck. When disruptions occur (e.g., port closures, geopolitical events, natural disasters), procurement teams lose 48–72 hours manually mapping out the "blast radius," sourcing alternative suppliers, and vetting them for ESG/compliance. By the time humans react, competitors have already consumed available alternative capacity.

ResilioChain is an event-driven, multi-agent AI squad that instantly reacts to supply chain disruptions. Powered by an advanced **GraphRAG (Graph Retrieval-Augmented Generation)** architecture, it autonomously maps the N-Tier impact of a disruption, identifies semantic supplier matches using **Voyage AI** embeddings, runs rigorous compliance checks, and presents a ready-to-execute mitigation brief to human operators in seconds—giving the enterprise a massive competitive advantage.

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2. Core Technology Stack

ResilioChain leverages a modern, agentic AI tech stack designed for speed, contextual accuracy, and deterministic outputs:

- **MongoDB Atlas (The Core):** Acts as the all-in-one operational, vector, graph, and agent-memory database.
- **Voyage AI:** Provides state-of-the-art embedding models and rerankers to process highly complex, domain-specific supply chain and compliance documents with unmatched semantic accuracy.
- **MCP (Model Context Protocol):** Standardizes the interface between the LLMs and MongoDB, exposing database queries and vector searches as secure, composable tools.
- **LangGraph:** The orchestration layer managing the multi-agent state machine and human-in-the-loop handoffs.
- **AWS Bedrock / AgentCore:** The secure, zero-retention LLM inference layer (Claude 3 Haiku/Sonnet) acting as the cognitive engine for the agents.

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3. Architecture Deep Dive: The GraphRAG Engine

ResilioChain doesn't just do basic RAG; it utilizes **GraphRAG**. Standard RAG fails in supply chains because it cannot understand deterministic relationships (e.g., Part A goes into Subassembly B). GraphRAG solves this by fusing MongoDB's graph traversal capabilities with Voyage AI-powered vector search.

A. The MongoDB Atlas & Voyage AI Layer (The Data Foundation)

MongoDB handles multiple distinct database workloads without the latency of zero-ETL pipelines, while Voyage AI ensures the unstructured data is perfectly vectorized.

- **M1: Supply Chain Graph (MCP Resource & Tool - `mcp: blast_radius`)**
  - *Function:* Calculates the N-Tier Deep Blast Radius (e.g., Lithium -> Battery Pack -> EV SUV -> Order #1044).
  - *Tech (The "Graph" in GraphRAG):* Uses MongoDB's native **`$graphLookup`** aggregation pipeline on a `depends_on` field to traverse the Bill of Materials (BOM) and instantly identify all downstream orders affected by a disrupted node.
- **M2: Sourcing Database (MCP Tool - `mcp: supplier_match`)**
  - *Function:* Identifies alternative suppliers globally based on capabilities, capacity, and unstructured historical performance data.
  - *Tech (The "RAG" in GraphRAG):* Supplier profiles and capabilities are embedded using **Voyage AI** (optimized for precise, domain-specific retrieval) and queried via **MongoDB Atlas Vector Search** for semantic matching rather than rigid keyword searches.
- **M3: Compliance Knowledge Base (MCP Tool - `mcp: compliance_check`)**
  - *Function:* Houses complex enterprise policies, ESG guidelines, and global sanction lists.
  - *Tech:* Utilizes **Voyage AI embeddings + reranking** combined with **MongoDB Vector RAG**. This ensures the Compliance Agent can accurately retrieve and reason over dense, legalese-heavy regulatory guardrails.
- **M4: LangGraph State & Memory (MCP Resource - `mcp://checkpoints/{thread_id}`)**
  - *Function:* Provides state persistence, auditability, and thread memory for the agent squad.
  - *Tech:* A standard MongoDB CRUD collection storing LangGraph checkpoint states, enabling "time-travel" debugging and human-in-the-loop approvals.

B. Model Context Protocol (MCP) Integration

To ensure the LLM layer interacts with MongoDB securely and reliably, ResilioChain utilizes the open standard **Model Context Protocol (MCP)**.

- **Standardized Tools:** The LLMs do not write raw database queries. Instead, Graph lookups and Voyage AI-powered vector searches are exposed as pre-defined MCP tools (`blast_radius`, `supplier_match`, `compliance_check`).
- **Context & Resources:** Thread memory and checkpoints are exposed as MCP resources, allowing the Supervisor Agent to read the exact state of the triage process at any time.

C. LangGraph Multi-Agent Orchestration Layer

ResilioChain utilizes a **Supervisor Agent pattern** deployed via LangGraph. The Supervisor routes tasks, maintains the global State Dictionary, and coordinates three specialized sub-agents:

1. **Impact Assessor Agent:** Triggered first. Uses the `blast_radius` tool (`$graphLookup`) to deterministically assess what internal products/orders are delayed.
2. **Sourcing Agent:** Takes the output from the Assessor. Uses the `supplier_match` tool (Voyage AI + Atlas Vector Search) to find viable backups.
3. **Compliance Agent:** Acts as the system's **Guardrail**. Takes the proposed backup suppliers and runs them against the `compliance_check` tool.

- *Self-Correction Loop:* If a supplier fails ESG/Sanctions checks, the Compliance Agent updates the state and kicks the flow back to the Sourcing Agent to find an alternative.

D. AWS Bedrock / AgentCore LLM Layer

- **Cognitive Engine:** Powered by Anthropic's Claude 3 family (Haiku for fast routing/tool calling, Sonnet for complex compliance reasoning) via AWS Bedrock.
- **Security & Compliance:** By using AWS Bedrock, ResilioChain ensures **Zero Data Retention**—sensitive supply chain data and supplier contracts are never used to train base models.
- **AgentCore Integration:** Acts as the MCP Client orchestrator, executing the prompts, reasoning over the state dictionary, and invoking the API calls back to the LangGraph/MongoDB layers.

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4. The Autonomous Workflow (Step-by-Step)

1. **Event Trigger:** An external alert (e.g., "Port of Baltimore Closed") is ingested.
2. **State Initialization:** LangGraph spins up a new thread, saving the initial state in MongoDB (M4).
3. **Blast Radius Calculation (Graph):** The Supervisor routes the alert to the **Impact Assessor Agent**, which traverses the supply chain graph in MongoDB (M1) to find affected SKUs.
4. **Alternative Sourcing (RAG):** The **Sourcing Agent** performs a semantic vector search powered by Voyage AI in MongoDB (M2) to find alternate logistics routes or part suppliers.
5. **Compliance Vetting (RAG):** The **Compliance Agent** runs the alternatives against internal ESG policies using Voyage AI embeddings + MongoDB Vector Search (M3).
6. **Resolution & Mitigation Brief:** Once a compliant alternative is found, the Supervisor Agent compiles the findings (Impact + Alternative + Cost Variance + Compliance Proof) into a Mitigation Brief.
7. **Human-in-the-loop:** The final brief is presented to the human operator for 1-click execution.

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5. Business Value & Hackathon Winning Differentiators

- **True GraphRAG Implementation:** Move beyond basic vector search by fusing deterministic graph traversals (`$graphLookup`) with probabilistic semantic search in a single pipeline, solving the exact reason standard RAG fails in supply chain use cases.
- **Best-in-Class Semantic Retrieval:** By utilizing **Voyage AI**, the system achieves industry-leading retrieval accuracy on dense, complex supply chain contracts and compliance policies, minimizing LLM hallucinations.
- **MCP First:** Showcases future-proof agentic architecture by decoupling database operations from LLM logic using the Model Context Protocol.
- **Radical ROI:** Reduces exception triage time from 72 hours to less than 30 seconds, allowing the enterprise to secure alternative supply chain capacity before market competitors.
- **Enterprise-Ready Security & Auditability:** Combines AWS Bedrock's zero-retention privacy with MongoDB's state checkpointing, ensuring every autonomous agent decision can be fully audited.

ResilioChain: The Live Demo Narrative

Scenario: The N-Tier Supply Chain Meltdown

The Setup (Addressing the Judges):
*"Judges, imagine you are the Global Logistics Director for an automotive enterprise. It's 3:00 AM on a Friday. Most of your team is asleep. Suddenly, news breaks: A massive localized flood has just shut down an industrial park in Dresden, Germany. Standard supply chain software just gives you a red flashing dot on a map. You have 72 hours of manual spreadsheet-digging ahead of you. Competitors are already dialing alternative suppliers. Let's see what happens when **ResilioChain** is on the clock."*

🚨 Step 1: The Trigger & State Initialization

- **The Action:** An automated webhook from a news/weather API drops an alert into the ResilioChain UI: `CRITICAL: Flooding at Dresden Industrial Park. Supplier 'Klausen Micro-Sensors' offline.`
- **Behind the Scenes:** The alert triggers the **LangGraph Orchestrator**.
- **The Tech Highlight:** Instantly, LangGraph initializes a new state dictionary. **MongoDB Atlas (M4 Collection)** logs this as a new thread via standard CRUD operations. We are now capturing every single agent thought and tool call for full enterprise auditability.

🕸️ Step 2: The "Graph" in GraphRAG (Blast Radius Calculation)

- **The Action:** The Supervisor Agent realizes it needs to know what this means for the business. It summons the **Impact Assessor Agent**.
- **Behind the Scenes:** The Agent invokes the MCP tool `mcp:blast_radius`.
- **The MongoDB Highlight:** *Standard RAG would fail here—you cannot do semantic search to figure out exact engineering dependencies.* Instead, ResilioChain uses **MongoDB's native `$graphLookup` (M1 Collection)**. In milliseconds, it traverses our complex Bill of Materials (BOM) graph:
  - *Klausen Micro-Sensors* ➡️ goes into ➡️ *Battery Management System (BMS)* ➡️ goes into ➡️ *EV SUV Battery Pack* ➡️ goes into ➡️ **High-Priority Fleet Order #1044 for a major enterprise client.**
- **The Impact:** In 2 seconds, ResilioChain knows exactly *who* is impacted and *what* parts are missing.

🔍 Step 3: The "RAG" in GraphRAG (Semantic Alternative Sourcing)

- **The Action:** We know what's broken; now we need to fix it. The Supervisor hands the state to the **Sourcing Agent** to find a replacement sensor.
- **Behind the Scenes:** The agent invokes the `mcp:supplier_match` tool.
- **The Voyage AI & MongoDB Highlight:** We aren't doing rigid keyword searches for "Klausen Sensor". We use **Voyage AI** to generate a highly precise, domain-specific embedding of the missing component's engineering specs. We pass this into **MongoDB Atlas Vector Search (M2 Collection)**.
- **The Result:** Atlas Vector Search instantly surfaces two alternative suppliers globally that have matching capabilities and active capacity: *Osaka Tech (Japan)* and *Shenzhen Innovate (China)*.

🛡️ Step 4: The Guardrail & Self-Correction (Compliance Vetting)

- **The Action:** We can't just blindly buy parts. The Supervisor passes the two options to our strict **Compliance Agent**.
- **Behind the Scenes:** The agent invokes the `mcp:compliance_check` tool. It takes the supplier profiles and runs them against our enterprise ESG (Environmental, Social, and Governance) policies stored in **MongoDB (M3 Collection)** using **Voyage AI RAG**.
- **The Agentic Magic (Crucial Hackathon Moment):**
  - The LLM reasoning engine (**AWS Bedrock / Claude 3 Sonnet** - acting with zero data retention for security) realizes that *Shenzhen Innovate* has a recent flag regarding localized labor compliance that violates our strict ESG rules.
  - The Compliance Agent *rejects* Shenzhen Innovate. It updates the LangGraph state, loops back, and successfully validates *Osaka Tech*.
  - **MongoDB Checkpoint:** Every step of this rejection and autonomous reasoning is saved in MongoDB M4, proving to compliance auditors *why* the AI made this choice.

Step 5: The Mitigation Brief & Human-in-the-Loop Hand-off

- **The Action:** Less than 15 seconds have passed since the flood alert. The UI pings the human operator.
- **The Result:** On the screen is a clean, generated **Mitigation Brief**:
  - **Incident:** Dresden Flooding affecting Klausen Sensors.
  - **Blast Radius:** Threatens EV Fleet Order #1044 (Calculated via `$graphLookup`).
  - **Mitigation:** Switch to Osaka Tech.
  - **Compliance:** Vetted and cleared against ESG policies (Via Voyage AI + Atlas Vector Search).
  - **Action Required:** `[CLICK HERE TO EXECUTE PURCHASE ORDER]`
- **The Wrap-up:** The human clicks "Approve".

The Pitch Conclusion (For the Judges)

"Judges, what you just saw wasn't just a chatbot. It was a deterministic, multi-agent squad solving a multi-million dollar problem in 15 seconds.

We didn't need three different databases to do this. ***MongoDB Atlas*** *was the absolute hero—acting as our operational database, our graph engine via* `$graphLookup`*, our vector store via Atlas Vector Search, and our LangGraph memory bank.*

By combining MongoDB with ***Voyage AI*** *for industry-best retrieval, orchestrating it with* ***LangGraph****, and keeping it perfectly secure with* ***MCP*** *and* ***AWS Bedrock****, ResilioChain delivers true* ***GraphRAG****. We didn't just find information; we reasoned over relationships, vetted alternatives, and saved the supply chain while the competition was still waking up. Thank you."*